

Evaluating Text-To-Music Generation

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Abstract

The rapid evolution of deep learning has driven significant progress in text-to-music generation. This report outlines the current research landscape and conducts a rigorous comparative evaluation of three distinct architectural approaches, benchmarking their capabilities across two complementary datasets.

1 Introduction

As a complex subfield of multimodal machine learning, text-to-music generation addresses the challenge of translating natural language descriptions into musically coherent auditory outputs that maintain strict semantic alignment with the input. This task presents unique difficulties, as models must not only capture the stylistic, emotional, and instrumental nuances dictated by the text, but also model the intricate, long-range temporal dependencies inherent in music.

1.1 Examples

To illustrate the practical application and the inherent complexity of this task, consider the following scenarios of text-to-music generation:

- **Example 1:**

Input Prompt: "A dark gothic orchestral soundtrack that begins with chaotic, dissonant strings and erratic percussion to evoke terror, before abruptly transitioning at the midpoint into a majestic, tragic waltz driven by a soaring choir and heroic brass."

Expected Output: The model must successfully synthesize an audio waveform capable of executing a drastic structural and emotional shift. Initially, it must generate dissonant acoustic textures and irregular rhythms to capture the requested horror atmosphere. Then, the model must demonstrate advanced temporal control by handling

the sudden transition into a structured, harmonious waltz rhythm, while seamlessly introducing a prominent vocal choir and majestic brass, maintaining strict semantic alignment with the evolving nature of the prompt.

(The input prompt references the 'Ludwig, the Accursed & Holy Blade' boss fight soundtrack from the video game Bloodborne and composed by Nobuyoshi Suzuki)

- **Example 2:**

Input Prompt: "A slow, melancholic solo piano composition featuring a delicate, nostalgic melody, gently underscored by the ambient sound of rhythmic ocean waves."

Expected Output: The model must seamlessly integrate two distinct acoustic modalities: melodic instrumental synthesis (the piano) and non-musical environmental textures (the waves). The system must exercise great control over dynamics and a slow tempo to accurately render the requested melancholic emotional nuance.

(The input prompt references the 'Dearly Beloved' title theme from the video game Kingdom Hearts and composed by Yoko Shimomura)

1.2 Real-world applications

Advancements in text-to-music generation have unlocked a diverse array of practical applications across multiple domains, such as generating background music for videos, podcasts, and social media content; producing jingles, soundscapes, and promotional tracks tailored for advertising; and serving as powerful co-creation tools for musicians, among numerous other emerging use cases.

2 Related Work

Earlier work by Dhariwal et al. [6] showed that combining multiscale VQ-VAEs for audio compression with large-scale auto-regressive transformers can generate high-fidelity, diverse songs with multi-minute coherence.

A subsequent important work by Kreuk et al. [12] introduced the first auto-regressive generative model for text-conditioned audio generation.

On the evaluation front, the two principal contributions come from Wu et al. [20] and Kilgour et al. [11]. Together, they introduced the state-of-the-art metrics for text-to-music models.

Most prominent contemporary models rely on the breakthroughs by Vaswani et al. [18], Rombach et al. [17], and Peebles et al. [16], relating to transformers, latent diffusion models, and diffusion transformers, respectively.

Building upon these foundational architectures, the field of music generation has seen rapid and significant advancements. Liu et al. [13] introduced AudioLDM, successfully adapting latent diffusion models for computationally efficient and high-quality text-to-audio generation. Copet et al. [4] presented MusicGen, demonstrating that a simple, single-stage auto-regressive transformer can achieve highly controllable and coherent music generation. Parallely, Google advanced the state of the art with MusicLM (Agostinelli et al. [1]) and their subsequent Lyria architecture, treating conditional music generation as a hierarchical sequence-to-sequence modeling task to produce high-fidelity audio from rich text descriptions. More recently, Evans et al. [8] released Stable Audio Open, further pushing the boundaries by providing an open-weights latent diffusion model optimized for generating variable-length stereo music and sound effects.

3 Datasets and Benchmarks

In this section, I consider the most significant and foundational datasets by classifying them into three categories.

3.1 Human-Annotated Datasets

One basic dataset is AudioSet, an expansive corpus of over two million annotated clips covering a wide array of general sound events. While AudioSet provides a massive scale for general audio research, it also served as the source material for a much more refined benchmark: MusicCaps

[1]. This highly specialized dataset extracts 5,521 music-specific clips from AudioSet and pairs them with extremely technical captions written by professional musicians. This level of expert annotation makes MusicCaps the undisputed gold standard for rigorous, objective model evaluation.

However, professional musical terminology does not always reflect how everyday users interact with generative AI. To bridge this gap, the Song Descriptor Dataset [14] (SDD) was introduced as a complementary benchmark. Unlike MusicCaps, SDD was created through a crowdsourcing initiative that asked ordinary people to describe the musical tracks.

3.2 Large Scale Datasets

Because a benchmark of 5,000 clips is vastly insufficient for training modern generative AI models from scratch, researchers repurposed older, massive datasets by utilizing LLMs to transform simple metadata tags into fully formed, descriptive natural language sentences.

Large-scale Pseudo-labeled MusicCaps [7] is an example of combining the previously mentioned music archives and prompting an LLM to write coherent, meaningful captions, ultimately generating a dataset large enough to support the heavy training requirements of modern text-to-music models.

Another key resource built on this approach is WavCaps [15]. This dataset contains approximately 400,000 audio clips paired with rich, ChatGPT-generated descriptions. This extensive collection is widely adopted to teach and refine the model’s precise text-to-audio alignment capabilities across a vast variety of sounds.

Similarly, the Free Music Archive [5] (FMA) serves as a foundational pillar within this category. Originally introduced as a massive open-source collection of over 100,000 tracks equipped with hierarchical metadata and genre tags, FMA was initially designed for music information retrieval tasks. However, much like the archives behind LP-MusicCaps, modern generative AI research has repurposed FMA by applying LLM-driven pseudo-labeling techniques.

Finally, the captioned version of MTG-Jamendo [2] addresses a distinctly different challenge in music generation. Unlike datasets composed of short audio snippets, MTG-Jamendo features full-length musical tracks useful for teaching AI the long-term structural progression, macro-

dynamics, and narrative evolution of a full musical piece.

3.3 Closed Source Datasets

While inaccessible to the public research community, proprietary datasets cannot be ignored, as they often define the commercial state-of-the-art and drive major industry breakthroughs.

The most prominent example is the proprietary dataset used to train Google’s MuLan [9] (Music-Language pre-training model). Ubiquitous across Google’s research papers, it represents a monumental achievement in data collection. To build it, Google extracted an astounding 44 million music clips directly from YouTube videos. To establish the crucial text-to-audio pairings required for training, they cleverly leveraged the natural language already present in the videos’ descriptions and user comments. This approach allowed them to create a massive, highly diverse dataset of real-world text-music associations that remains unparalleled in scale.

4 Existing Tools, Libraries, Papers with Code

In this section, there is an overview of the most prominent tools, software libraries, and official code repositories that facilitated my comparative evaluation of text-to-music generation models.

4.1 Existing Tools

In today’s machine learning environment, the Hugging Face [19] platform emerges as a key and highly robust resource for both researchers and developers. By offering a unified API and a vast library of pre-trained model weights along with open-source datasets, Hugging Face removes the necessity of constructing data ingestion and inference pipelines from various sources manually.

4.2 Libraries

To implement the practical aspects of my research, I utilized various specialized Python libraries, which can be categorized based on their function in the workflow.

Data Management

Working with audio processing involves efficiently managing substantial waveforms and associated metadata. I used the `datasets` and `soundfile` libraries to retrieve, store, and decode the audio files.

Inference

For the generative models, I relied on the `diffusers` and `transformers` libraries.

The diffusion-based baseline, AudioLDM [13], is instantiated through the first library, also using half-precision to reduce GPU memory consumption.

In contrast, the auto-regressive approach, MusicGen [4], is implemented with the second library and requires both an `AutoProcessor` for text tokenization and the pretrained model for audio generation.

Finally, the diffusion transformer architecture, Stable Audio Open [8], is also loaded via `diffusers`. Much like AudioLDM, it utilizes half-precision to optimize hardware resources, but operates natively on a stereo latent space to generate high-fidelity, dual-channel audio outputs.

Evaluation

To extract objective evaluation metrics, I relied on a combination of specialized libraries and custom helper functions.

For the CLAP score [20], I used the `ClapModel` and `ClapProcessor` components provided by the `transformers` library.

For the computation of the FAD [11], several external toolkits can be used, including `fadtk` and the official `frechet_audio_distance` by Google Research. Since my evaluation is based on the FMA dataset, I implemented a set of custom helper functions to adapt the FAD computation pipeline to my specific experimental setting.

4.3 Papers with Code

The scientific articles and corresponding source code repositories associated with the aforementioned pipelines are publicly accessible and extensively documented. Among the principal references defining the current open-source state of the art, and forming the foundation of the methodologies adopted in this work, are the following:

- **AudioLDM:** “*AudioLDM: Text-to-Audio Generation with Latent Diffusion Models*” by Liu et al. (2023) [13].
- **MusicGen:** “*Simple and Controllable Music Generation*” by Copet et al. (2023) [4].
- **Stable Audio Open:** “*Stable Audio Open*” by Evans et al. (2024) [8].

- **CLAP:** “*Large-scale Contrastive Language-Audio Pretraining with Feature Fusion and Keyword-to-Caption Augmentation*” by Wu et al. (2023) [20].
- **FAD:** “*Fréchet Audio Distance: A Reference-Free Metric for Evaluating Music Enhancement Algorithms*” by Kilgour et al. (2019) [11].

5 State-of-the-art Evaluation

Current state-of-the-art evaluation protocols in the literature are generally based on a hybrid assessment framework, combining objective quantitative metrics with subjective human evaluations. Objective measures are employed to assess acoustic quality and semantic consistency between text and audio, whereas human judgments are used to evaluate perceptual aspects such as musicality, coherence, and emotional expressiveness.

5.1 Objective Metrics

One widely adopted metric is the CLAP [20] (Contrastive Language-Audio Pre-training) score, which measures the degree of alignment between the generated audio and the input text prompt. A dedicated CLAP model encodes both the audio and the textual description into a shared embedding space and computes a similarity score (typically cosine similarity) between the resulting embeddings, providing a quantitative estimate of how well the two correspond.

FAD [11] (Fréchet Audio Distance) is another key evaluation metric used to assess the quality of generated audio by measuring the distributional distance between synthetic and real audio samples. It is based on the Fréchet distance between multivariate Gaussian distributions fitted to audio embeddings. Specifically, audio signals are first passed through a pretrained embedding network, which maps each audio clip into a high-level feature representation. The sets of embeddings for both real and generated audio are then modeled as Gaussian distributions, characterized by their respective means and covariances, from which the Fréchet distance between the two distributions is computed.

The final objective metric is the RTF (Real-Time Factor), which measures the computational efficiency of the method in terms of the ratio between the time required to generate an audio clip and its actual duration.

5.2 Subjective Evaluation

Researchers widely recognize that objective metrics such as FAD and CLAP cannot fully capture the richness of musical perception. A generated piece may exhibit excellent performance in terms of distributional similarity to real audio (low FAD) and strong alignment with the input text prompt (high CLAP), yet still fail to deliver a coherent musical structure, exhibiting issues such as harmonic dissonance, weak rhythmic organization, or limited emotional expressiveness.

For this reason, the most comprehensive evaluation standard for generative music remains the Mean Opinion Score [10] (MOS), which is entirely based on human judgment. The standard MOS protocol typically proceeds as follows:

1. A heterogeneous group of evaluators is recruited, including both trained musicians and non-expert listeners.
2. Participants take part in blind listening experiments involving a set of generated audio samples.
3. Each sample is rated on a Likert scale from 1 to 5, with respect to subjective criteria such as overall musical quality, perceived pleasantness, and audio fidelity.

However, subjective evaluation also presents several limitations. First, evaluation inconsistency arises because different listeners may interpret and score the same audio differently, leading to significant inter-rater variability. Second, evaluation complexity is inherent to the multidimensional nature of music perception, where aesthetic qualities such as harmony, rhythm, and emotional expressiveness are difficult to jointly assess in a consistent and exhaustive manner. Finally, time and cost constraints represent a major bottleneck, as large-scale listening studies require substantial human effort and resources and often suffer from limited scalability and reproducibility. [3]

6 Comparative Evaluation

This section will describe what actually takes place within the provided Jupyter Notebook.

As mentioned earlier, I rely on various external libraries to conduct the comparative analysis. In addition to these, I have developed a set of custom helper functions, which I have organized into three categories for greater clarity:

- **Visualization:** The two functions in question are used to show playable audio examples wrapped in a nice HTML div.
- **Metrics Computation:** Three functions are dedicated to computing the CLAP and FAD metrics.
- **Reproducibility function:** A final function is used to apply a global state that ensures the reproducibility of the evaluations and experiments.

There is also a small section that allows you to modify the number of samples to retrieve from the datasets, which is also the number of samples the tested models will generate; the number of samples you want to display using the visualization helper functions (which you can then choose to play); and finally, the seed for reproducibility.

In the fourth section, the datasets I rely on, MusicCaps and FMA (already described in Section 3), are downloaded. Since FMA does not have actual captions like MusicCaps does, I decided to generate them synthetically using the 'genre' metadata. This way, we have extremely complex prompts (those from MusicCaps) and extremely simple prompts (the synthetic ones produced from FMA).

In the fifth section, I simply show examples from both datasets and their corresponding captions.

Next, we reach the section dedicated to the baseline model, AudioLDM v1 (full). This latent diffusion model is tasked with generating a total of 400 five-second music tracks: 200 based on the MusicCaps captions and 200 based on the synthetic FMA prompts. Naturally, once the generation process is complete, this same section presents a few sample outputs for both datasets.

Next is the MusicGen (small) approach, which, in contrast, is an autoregressive model. Similarly, it is tasked with generating 400 tracks equally divided between the two datasets, and once again, a few examples are provided for listening.

The final approach is Stable Audio Open, which relies on a diffusion transformer architecture. Just like the previous methodologies, it is tasked with generating 400 tracks equally divided between the two datasets, and as before, a selection of generated samples is provided for listening.

The eighth section focuses on evaluating the generated tracks using the CLAP score, previously

introduced in Section 5. In short, CLAP functions much like CLIP does for images: it projects both the audio and text prompt embeddings into a shared latent space to determine their cosine similarity.

Finally, there is an evaluation section for the FAD and another evaluation section for the RTF, both of which were also described in Section 5.

6.1 Results & Discussion

When analyzing the CLAP scores, it is essential to first premise that, unlike CLIP, the inherently subjective nature of music leads to significantly lower average scores, making the metric somewhat less reliable, as previously discussed. Nevertheless, a higher value still indicates a stronger correlation between the text and the generated audio. In this regard, Stable Audio Open outperforms its competitors across both prompt types, reaching a peak score of 0.4407 with the detailed prompts from MusicCaps.

Dataset	Model	Avg. CLAP Score
MusicCaps	AudioLDM	0.230300
MusicCaps	MusicGen	0.372300
MusicCaps	Stable Audio	0.440700
FMA	AudioLDM	0.224000
FMA	MusicGen	0.315300
FMA	Stable Audio	0.396300

The evaluation of the FAD score requires a more nuanced interpretation in this study. Typically, FAD compares generated audio against a high-quality background dataset, meaning that a lower score indicates superior acoustic fidelity. However, for this experiment, I deliberately selected the FMA dataset as the reference dataset. Because FMA consists of downsampled audio optimized for efficiency, the standard logic is inverted: a higher FAD score actually signifies better audio quality, as it represents a greater mathematical distance from a lower-quality acoustic distribution. This tailored methodological approach yields a highly insightful finding: it exposes a distinct limitation in AudioLDM, which tends to generate tracks with noticeable background noise, a flaw that is far less prominent in MusicGen and Stable Audio Open.

It is important to acknowledge that this approach is not strictly comparable to the original methodology. However, this was a deliberate choice aimed at introducing an additional experi-

mental dimension, and the overall results nonetheless appear to be concordant.

As demonstrated in the table, the highest quality is achieved by MusicGen when utilizing the extremely simple synthetic prompts derived from the FMA dataset. This is by no means a coincidence. Evidence suggests that both models, albeit to vastly different extents, suffer from noise hallucinations when confronted with overly detailed prompts cluttered with stop words.

Dataset	Model	FAD
MusicCaps	AudioLDM	0.416600
MusicCaps	MusicGen	0.688600
MusicCaps	Stable Audio	0.613500
FMA	AudioLDM	0.467700
FMA	MusicGen	0.791000
FMA	Stable Audio	0.600100

Naturally, the superior audio quality and stronger text alignment achieved by Stable Audio Open and MusicGen come at a cost. This trade-off is evident in their elevated RTF; a value greater than 1.0 indicates that the models are unable to generate audio in real-time, unlike the AudioLDM baseline. This limitation could have significant implications for certain practical applications. Nevertheless, from a subjective evaluation standpoint, the final output produced by the two methods vastly outshines the baseline, extending far beyond what quantitative metrics like FAD and CLAP can capture. Therefore, I consider this computational cost to be more than justified.

Model	RTF (s)
AudioLDM v1	0.727100
MusicGen Small	1.193400
Stable Audio Open	1.456200

7 Conclusions

This report summarizes the state-of-the-art in text-to-music generation from various perspectives, examining three notable models with distinct architectures. A key direction for future research involves implementing techniques to ensure greater temporal coherence, with the goal of overcoming the current generation limit, which is presently capped at 30 seconds for MusicGen and at 47 seconds for Stable Audio Open. Furthermore, applying text augmentation techniques would be highly

beneficial to enhance the models' overall performance.

Finally, it is worth noting that currently available datasets often lack clarity and comprehensive metadata. Therefore, it would be interesting to develop specialized models focused on specific musical genres. This approach would help mitigate the hallucinations triggered by overly descriptive text prompts and significantly enhance the reliability of evaluation metrics.

Use of GenAI and LLM tools. Here, I will detail my personal use of Large Language Models and other AI-based tools, referencing them systematically according to the structure of the paper.

The initial phase of AI usage was dedicated to gathering information and establishing a structural outline for the research. For this purpose, I initially employed Google Gemini and Elicit.

Furthermore, I utilized tools such as ChatGPT, Google Gemini, and Grammarly primarily for text rephrasing and to expedite the proofreading and error correction processes.

The input prompts for Examples 1 and 2 in the introduction were generated entirely using ChatGPT, with the aim of ensuring maximum objectivity.

Additionally, I occasionally utilized Google Gemini to expedite the retrieval of links to relevant research papers.

Google Gemini was also employed to troubleshoot and resolve issues or library dependency conflicts within the code of the project.

Finally, I relied on ChatGPT to enhance the visual presentation of the project notebook, specifically by structuring the code with well-formatted Markdown cells enriched with emojis.

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